Name – Arpita Waghmare

PRN-24070521006

Batch-A1 SEC-A

Open

postfi

```
%{
#include <stdio.h>
#include "y.tab.h"
%}

%%
[0-9]+ {yylval = atoi(yytext);return NUMBER;}
[+*] {return yytext[0];}
[ \t] {;}
\n {return '\n';}
. {printf("Invalid Expression: ");}
%%
int yywrap(void)
{
    return 1;
}
```

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Batch-A1 SEC-A

Open postl ~/Desktop

postfix.y

```
%{
#include <stdio.h>
#include <stdlib.h>
int yylex();
void yyerror(char *s);
%}
%token NUMBER
%%
stmt: expr '\n'   { printf("Result: %d\n", $1); }
|
    ;
expr:NUMBER         { $$ = $1; }
   | expr expr '+'   { $$ = $1 + $2; }
   | expr expr '*'   { $$ = $1 * $2; }
   ;
%%
void yyerror(char *s)
{
   printf("Error");
   exit(0);
}
int main(void)
{
   yyparse();
   return 0;
}

```

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Batch-A1 SEC-A

```
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ yacc -d postfix.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ flex postfix.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ gcc y.tab.c lex.yy.c -ll
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ ./parser
2 3 +
Result: 5
^Z
[3]+  Stopped                 ./parser
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ yacc -d postfix.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ flex postfix.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ gcc y.tab.c lex.yy.c -ll
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$ ./parser
2 + 3
Error: syntax error
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/Postfix$
```

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Batch-A1 SEC-A

```
%{
#include "parser.tab.h"
%}

%%

[0-9]+    { yylval = atoi(yytext); return NUMBER; }
[+*^]     { return yytext[0]; }
[ \t]     { }
\n        { return '\n'; }
.         { printf("Invalid Expression: "); }

%%

int yywrap(void)
{
    return 1;
}
```

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Batch-A1 SEC-A

```
%{
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int yylex(void);
void yyerror(char *s);
%}

%token NUMBER

%%

stmt: expr '\n'
      { printf("Result: %d\n", $1); }
    ;

expr: NUMBER
      { $$ = $1; }
    | expr expr '+'
      { $$ = $1 + $2; }
    | expr expr '*'
      { $$ = $1 * $2; }
    | expr expr '^'
      { $$ = (int)pow($1, $2); }
    ;

%%

void yyerror(char *s)
{
    printf("Error: syntax error\n");
    exit(0);
}

int main(void)
{
    yyparse();
    return 0;
}
```

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PRN-24070521006

Batch-A1 SEC-A

```
ayush@Arpita UCRT64 ~
$ cd /c/Users/ayush/OneDrive/Desktop/Postfix

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ ls
lex.l  parser.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ bison -d parser.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ flex lex.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ gcc parser.tab.c lex.yy.c -o parser.exe
lex.l:2:10: fatal error: y.tab.h: No such file or directory
    2 | #include "y.tab.h"
      |          ^~~~~~~~
compilation terminated.

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ ^C

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ bison -d parser.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ flex lex.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
$ gcc parser.tab.c lex.yy.c -o parser.exe

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Postfix
2 3 ^
Result: 8
```